

Wenyu Chiou

AI Agent Infrastructure & Evaluation Engineer

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I build AI-agent and research infrastructure — MCP pipelines, multi-agent orchestration, and agent harnesses — and I study when LLM agents can be trusted to stand in for real people, using behavioral ground-truth data I collected myself.

— EXPERIENCE

Graduate Research Assistant — Complex Adaptive Water Systems Lab, Lehigh University Aug 2024 – present

- Lead an **LLM-agent validity** program — whether LLM agents reproduce real human decisions — tested against a **937-household survey I designed and fielded**; co-authored the published *Water Resources Research* (2026) study that substitutes LLMs as the behavioral engine of the simulation.
- Building **WAGF**, a governance framework that constrains LLM agents with domain rules and behavioral theory so coupled-simulation outcomes stay stable and physically sensible (in preparation for release).
- Built **FLOODABM**, a **52,141-household** coupled agent-based × catastrophe simulation with NFIP mechanics, validated against OpenFEMA insurance-claims data.

Independent Open-Source AI Infrastructure 2024 – present

- Built **research-hub**, an **MCP server + CLI** that turns Zotero, Obsidian, and NotebookLM into an AI-operable research workspace: searches arXiv / Semantic Scholar / PubMed / CrossRef, ranks with BM25, and gates retrieval quality with **recall@k evaluation against golden fixtures**. **PyPI research-hub-pipeline v1.1.1**; listed in **Awesome MCP Servers**; CI across **3 OS × 4 Python versions**.
- Built a multi-agent tooling stack — **ai-research-skills** (15-skill research catalog), **fable-method-harness** (cost-routing + audit-trail harness), and **agent-collab-skills + codex-delegate** (orchestration + delegation) — on the **OpenAI and Anthropic APIs, MCP, and Ollama**, with anti-fabrication contracts verified from git state and CI on Ubuntu and Windows.
- **10 merged pull requests** into third-party open-source projects, including a root-caused engine-selection fix with a regression test that fails on main (BuilderIO/agent-native ×4) and a Windows file-permissions security fix (langchain-ai/openwiki).
- **awesome-agentic-ai-zh** — a trilingual agentic-AI learning curriculum with CI-enforced locale parity; **4.5k+ GitHub stars**, 588 forks.

— PUBLICATION

Yang, Y. C. E., & Chiou, W. (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111.

doi.org/10.1029/2025WR042111 Published

— EDUCATION

Ph.D., Civil & Environmental Engineering, Lehigh University 2024 – present · expected 2028

M.S., Hydrological and Oceanic Sciences, National Central University Sep 2021 – Jun 2023

B.S., Earth Sciences, National Central University Sep 2017 – Jun 2021

— SKILLS

LLM & agent systems: OpenAI & Anthropic APIs · Model Context Protocol (MCP) · multi-agent orchestration · agent governance & validation · retrieval (BM25) + recall@k evaluation · Ollama (local models)

Research automation: Zotero · Obsidian · NotebookLM · arXiv / Semantic Scholar / PubMed / CrossRef

Programming & tooling: Python · pytest + property-based testing · CI (GitHub Actions) · Git · cross-platform (Windows / POSIX)

Modeling & risk analytics: structural equation modeling (SEM) · Bayesian calibration · NFIP mechanics · exceedance-probability curves · coupled ABM × catastrophe modeling